Ashwin Rajan R

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Summary

Backend and cloud infrastructure engineer with hands-on AWS experience building scalable APIs, distributed systems, CI/CD pipelines, and production-grade cloud deployments.

Technical Skills

Languages: Python, Java, JavaScript/Node.js, C++, SQL, Bash

Backend: FastAPI, REST APIs, OpenAPI/Swagger, SQLAlchemy, Celery, Nginx, WebSockets

Cloud & DevOps: AWS (EC2, ECS, S3), Azure, Docker, GitHub Actions, Terraform

Data & Messaging: PostgreSQL, Redis, Kafka, MinIO/S3

Systems: Linux, Git, CI/CD, Networking, TCP/IP, DNS, VPC, pytest

Experience

ZettaOne — Software Engineer Intern Jun 2026 – Present
Backend / Cloud Infrastructure *Bengaluru, India*

- Build AWS-based ingestion pipelines for IoT telemetry using Python and Docker to process edge-device data in near-real time.
- Profile pipeline performance to identify bottlenecks, and deploy FastAPI inference services in containerized AWS environments.

Projects

TrustLens — Explainable Fraud Detection Platform GitHub 2026

- Built an explainable fraud-detection backend in FastAPI with async SQLAlchemy and PostgreSQL, structured as a bounded modular monolith with per-domain service interfaces.
- Designed an idempotent Celery + Redis pipeline (OCR → fraud analysis → forensics → risk scoring) backed by a transactional-outbox event log (Kafka in prod) for durable, replayable processing.
- Secured the platform with JWT authentication, Fernet-encrypted PII, and an immutable WORM audit trail with a tamper-evident hash chain; containerized via Docker Compose with GitHub Actions CI.
- Added a NetworkX entity-relationship graph to detect shared-identity networks, mule accounts, and fraud rings, plus a governed ML second-opinion model with SHAP explanations and KS-test drift detection.

Flimo — Semantic Search Platform GitHub 2025

- Cut deployment time to 68 seconds by building a GitHub Actions CI/CD pipeline (push → SSH → docker build → health check).
- Lowered infrastructure cost to approximately $34/month by serving HTTPS through a Cloudflare Tunnel running as a systemd service with auto-restart.
- Deployed FastAPI and FAISS-based semantic search on AWS EC2 (t3.medium) behind an Nginx reverse proxy, using a multi-stage Dockerfile.
- Built a hybrid personalization engine generating composite taste vectors from user view history, filtered by rating and popularity thresholds.

FedLearn — Browser-Native Federated Learning Platform GitHub 2026

- Developed a browser-native federated learning platform with isolated multi-tenant training jobs and REST APIs for on-the-fly job configuration.
- Reduced tensor serialization latency from ∼45ms to under 1ms using a zero-copy binary transport protocol replacing JSON.
- Cut network utilization 10–100× via Top-K gradient sparsification and INT8 quantization, and tolerated ∼30–40% adversarial workers using Weiszfeld geometric-median aggregation.
- Eliminated straggler bottlenecks with asynchronous FedBuff aggregation and FedProx regularization for non-IID stability; added worker-side differential privacy (L2 clipping + Gaussian noise) tracked by a Rényi DP accountant.

Education

Vellore Institute of Technology, Bhopal Sep 2023 – May 2027
Bachelor of Technology in Computer Science and Engineering *CGPA: 8.36/10*

- Relevant Coursework: Operating Systems, Computer Networks, Distributed Systems, Cloud Computing, DevOps

Achievements & Certifications

- Solved 200+ Data Structures and Algorithms problems across coding platforms.
- Ranked in the Top 100 at the Zelestra x AWS ML Ascend Challenge.
- AWS Certified Cloud Practitioner (2026).
- Applied Machine Learning in Python Certification (2024).